TEDDY GARZA

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EDUCATION

The University of Texas at Austin Expected May 2029
Bachelor of Science, Mechanical Engineering
GPA: 3.96/4.0

EXPERIENCE

Intuitive Machines | *Structures Design Intern* June 2026 - Aug 2026

- Reduced Laser Retroreflector Array payload bracket mass by 61% at a 2.5 safety factor under 60g static load case in Creo
- Redesigned LEIA payload sunshade over 7 modal-analysis iterations into a structure 3.2x stiffer and 4% lighter
- Built a full-scale IM-4 communications box mockup used for component placement and harness routing on the flight vehicle
- Ran tensile tests and reduced force-displacement data in Excel to find composite Young's modulus, Poisson's ratio, and CTE

Longhorn Racing Electric (Formula SAE)

Front Unsprung (Wheel Assembly) Lead *Aug 2026 - Present*

- Own front wheel assembly packaging, setting up the wireframes and parametric SolidWorks documents the team builds from
- Ran Ansys FEA and simulations to trace sources of toe compliance and help spec a new compliance budget for stiffer parts
- Interface with suspension and brakes teams to coordinate packaging and design constraints across subsystems

Vehicle Dynamics Member *Sept 2025 - Aug 2026*

- Sized wheel bearings in MESYS to cut unsprung mass while increasing stiffness, reducing toe compliance during cornering
- Designed 6 unique critical wheel assembly components in SolidWorks to meet strict FSAE packaging constraints including custom front and rear hubs, centerlocking nut, and magnet-integrated bearing preload nuts
- Validated structural integrity and optimized mass using Ansys Workbench FEA to simulate static load distribution
- Fabricated precision components using Fusion 360 CAM toolpaths, a 3-axis CNC mill, and manual lathes
- Led the mechanical integration of a wheel speed sensing system, designing a custom IP-67 rated enclosure

Space Center Houston | *Guest Operations Associate* Nov 2024 - Aug 2025

- Managed high-volume guest flow and quickly troubleshot bottlenecks during peak attendance hours
- Resolved guest logistical issues and accurately operated point-of-sale systems to maintain efficient daily admissions

NASA Johnson Space Center | *Robotics Engineering Mentee* Aug 2024 - Dec 2024

Mentor: Leslie Quiocho, Senior Robotics Engineer

- Programmed C++ firmware for an arduino to process real-time accelerometer data and calculate vehicle stopping distance using kinematic equations to predict crash severity and optimize airbag deployment
- Communicated technical progress and findings through regular discussions and design reviews with a mentor

LEADERSHIP & ACTIVITIES

CCISD Superintendent's Student Advisory Council (3 years)

- Selected as 1 of 24 student representatives for a district of 40,000+ students, who advocated for student needs and successfully implemented changes to the district-wide GPA policy

CBHS Science Fair Club President (2 years)

- Revitalized the campus science fair program by founding a dedicated club and launching a summer workshop that trained 50+ students in scientific methodology and experimental design

Other Leadership: Principal's Forum Member (3 years), CCISD School Safety and Security Committee Student Representative (1 year)

STEM Endeavors: Robotics Competitions; VEX, BEST, FRC (7 years), Systems Integration Lead (2 years), UIL Science Team (1 year) Science and Engineering Fair Participant (13 years)

Collegiate Organizations: Longhorn Racing Electric (Formula SAE Electric), American Society of Mechanical Engineers

PROJECTS

“The Innovation of a Robotic Hand Integrated with a Haptic Glove” Aug 2023 - Apr 2024

AP Capstone Research Project

- Individually designed and constructed a robotic hand controlled by a haptic glove, aiming to provide users with enhanced dexterity and reduced risk in hazardous fields
- Leveraged technical skills, including OnShape, 3D printing, C++, and soldering, to bring the project to fruition
- Authored a research paper detailing the project's background research, design, methodology, and findings

SKILLS

Technical: SolidWorks, Creo, Creo Simulate, Excel, Ansys Workbench, OnShape, Fusion360 CAM, Mesys
Certifications: Machine Shop Certification
Eligible to work in the U.S. with no restrictions: U.S. Citizen